

Avinash Agarwal

Civilization, AI, Healthspan, Cosmos

Mathematics | Engineering | Superintelligence | Systems

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SUMMARY

Engineering veteran with **8+ years of industry leading experience at Meta and Microsoft**, passionate about tackling civilizational challenges, especially Healthspan and Space exploration! Proven track record of architecting and scaling highly available distributed systems processing billions of requests, leading cross-functional teams, and driving multi-million-dollar cost optimizations.

EXPERIENCES

∞ Meta

August 2025 — Present

Full-time (supervised by [Gary G. Liu](#) & [Bill Hamilton](#))

Menlo Park, California, USA

- **Applied AI & LLM Evaluations:** [Drafted into the Applied AI organization](#) to create model-evaluation tasks for Meta's next-gen LLM codenamed Avocado, under Meta's Superintelligence Labs.
- **AI for Healthspan:** Authored a visionary project proposal within the Applied AI org to process vast genetic datasets; Advancing milestones like Google's AlphaFold, the proposal leverages modern AI to establish novel health benchmarks and accelerate cures for aging, rare-diseases, etc.
- **Engineering Productivity:** Root-caused an issue around pull-request insights where reviewer delays were incorrectly and entirely attributed to authors; filed a high-priority bug to ensure accuracy and fairness in developer analytics for company-wide engineering productivity.
- **Stability Lead:** Championed platform reliability operations for the unified authentication infrastructure, by actively triaging and resolving critical alerts, identifying and filing tasks for key SLI/SLO coverage gaps, systemic propagation delays around alert resolution, etc.
- **WhatsApp-based Authentication:** Authored a technical proposal to prioritize secure code-send channels while projecting multi-million dollars in cost-savings; Fixed critical bugs around contact eligibility and internal server semantics; Triaged further security improvements, video-call discrepancies, etc.
- **Meta.ai Platform Reliability:** Stabilized login flows for the Meta.ai application by systematically resolving complex, intermittently failing cross-platform test suites, significantly improving CI/CD reliability.
- **Automated On-Call Agent:** Enhanced an alert summarizer agent by resolving prompt completeness errors; Collaborated directly with the company-wide on-call platform owners to include this agent in their workflow, significantly improving engineering productivity.
- **On-Call Operations & Incident Management:** Completed specialized onboarding; monitored dashboards, triaged alerts, completed reports, proactively root-caused a privacy anomaly, and collaborated with relevant owners to improve time-to-mitigate and prevent future outages.
- **Secured the runner-up position in an internal Hackathon:** Demonstrated high engineering velocity by rapidly fixing a high volume of issues while triaging improvements for the hackathon platform.
- **Global Identity Architecture & Execution:** Led the end-to-end design and implementation of 8+ distinct account recovery eligibility methods; Instrumented relevant telemetry validating the safe transitioning of billions of requests across the Meta ecosystem to a unified authentication infrastructure.
- **Observability & Quota Management:** Audited telemetry sources to resolve persistent capacity issues; optimized sampling rates and created distinct logging partitions for partner teams.
- **Privacy-Safe Infrastructure:** Modernized legacy components to use privacy-safe wrappers while onboarding to the unified authentication infrastructure team.
- **Infrastructure Bootcamp:** Dedicated almost a month gaining hands-on training to successfully onboard into my officially hired role on **Infrastructure Engineering** at Meta.



○ Azure Resource Graph

- **Championed increasing partition-key cardinality**, leveraging data-driven demonstrations to root-cause and resolve systemic skewness across the entire stack.
- **High-Performance (de)serialization**: Independently root-caused bottlenecks via profiler analysis and championed the stack-wide adoption of `System.Text.Json`, leveraging its faster, lower-allocation APIs to optimize payload processing at scale.
- **Memory Optimization**: Championed selective caching of internal columns informed by customer query telemetry to significantly reduce active memory footprint.
- **Engineered data ingestion components** hosted on the Azure Synapse platform to support dynamic time-window loading for security datasets.
- **SQL Query Optimization**: Optimized complex queries executed on Azure Synapse to efficiently load and merge multiple continuous data streams.
- **Profiled and optimized services** eliminating contention bottlenecks and achieving **nearly 10%** improvement in compute, memory and network efficiency.
- **On-Call Efficiency**: Improved on-call operations with multiple enhancements to internal monitors, documentations, etc.

○ Azure Resource Change Analysis

- **Backend General Availability**: Deployed relevant services, dependencies, etc. across US and China Government clouds, partnered with authorized personnel for sovereign air-gapped environments, achieving critical compliance milestones directly supporting Azure's enterprise market positioning.
- **Partition Skewness Reduction**: Optimized data-ingestion pipelines, achieving over 30% improvement in compute, memory and network efficiency.
- **Architectural Evolution**: Championed metadata enhancements and optimizations to the Change Analysis payload with extensive collaboration across teams and disciplines.
- **Change Actor General Availability**: Delivered a new compliance and auditing functionality enabling customers to audit WHO initiated a change and WHICH client interface (Portal, CLI, SDK, etc.) was used to make these changes.
- **Front-End General Availability**: Worked on improving [performance Lighthouse](#) scores, deployments and test automation, completing accessibility fixes, etc.
- **Change Safety Collaboration**: Provided technical guidance to a partner team around effectively integrating and referencing Change Analysis resources within their workflows.
- **Streaming Pipeline (1B+ daily notifications) Migration**: Authored an approved architectural design-doc and drove the foundational execution steps for migrating the high-throughput streaming data pipeline to its dedicated scale unit.



○ Scaling for a Large Enterprise Customer (1B+ course completions)

- **Performance Profiling & Load Testing**: Performed intensive load testing and performance profiling using Apache JMeter, ensuring platform stability for massive-scale deployments.
- Identified and contributed a fix ([GitHub Issue #46](#)) for [Azure Load Testing](#) + [Terraform](#) setup, effectively adding Azure-Load-Testing to the [multi-cloud cohort supported by Terraform](#);
- **System Optimization & Throughput**: Optimized core services to achieve a **20x throughput improvement** by studying and refining SQL query execution plans, eliminating redundant network calls, and implementing strategic caching layers.
- **High-Scale Data Migration**: Architected APIs to execute the migration of 160M+ course-completion records from another major cloud provider to Azure ensuring data integrity during the transition.
- **Technical Leadership & Design**: Championed an extensible architecture that minimized long-term technical debt, while balancing rapid iteration requirements.

- **UNICEF Learning Passport (powered by Microsoft Community Training)**
 - **Offline Innovation (Hackathon Lead):** Spearheaded a hackathon project to host the entire cloud service locally on Windows machines. The platform was recognized as one of **TIME's Best Inventions of 2021** for providing continuous education to children in conflict zones and refugee settings.
 - **Global Societal Impact:** This "offline-first" prototype was the key technical catalyst that prompted UNICEF to launch the initiative using Microsoft Community Training for no-connectivity regions, contributing to a high-scale initiative that reached **7.4 million learners across 43 countries** by 2024.
 - **Strategic Architectural Leadership:** Championed and worked on containerization and migration to cross-platform .NET Core; this move was critical for the platform's ability to run on evolving hardware in emerging markets.
 - **Operational Support & Edge Strategy:** Worked on **Azure IoT Hub** integration to support remote management and observability for offline learning devices; assisted in testing data-sync to ensure reliable curriculum delivery in offline settings.
- **Swachh Bharat Mission (powered by Project Sangam)**
 - Directed an engineering team through the end-to-end implementation of key mobile features, culminating in the successful launch of the official **Swachh Bharat Mission eLearning mobile app**.
 - Spearheaded the rapid triage of launch-critical issues in close collaboration with **Aditya Agarwal** and product management teams to ensure a smooth migration of the training platform.
 - Architected and implemented a high-impact search-enabled dropdown to resolve a critical UI bug; this allowed users to efficiently navigate and select from **the hundreds of cities where the platform was deployed**.
 - **Contributed to a high-scale social impact project** that successfully trained and empowered over **110,000+ municipal functionaries** across **4,000+ cities** in India.
- **Platform Localization**
 - Initially worked on comprehensive support for **right-to-left (RTL) formats** before assuming full platform localization ownership.
 - Spearheaded global accessibility by **engineering an internal localization framework** and integrating with manual translation pipelines.
 - **Live Demonstration:** Presented RTL capabilities to a large audience, gracefully utilizing live product errors to demonstrate comprehensive, end-to-end localization coverage.
 - Conducted a **comprehensive investigation on ligature support** to ensure accurate rendering of complex scripts across the ecosystem.
 - **Custom PDF Certification Engine:** Architected a templated HTML-to-PDF generation feature by defining rigorous localization and security requirements; strategically collaborated with IronPDF to accelerate time-to-market.
- **BCDR & Global Traffic Management:** Designed and Implemented a mission-critical Business Continuity and Disaster Recovery architecture using **Azure Front Door** (already benchmarked by **LinkedIn**), significantly improving page-load latencies, establishing a resilient global load-balancing layer and hardening the platform's security posture for massive-scale international traffic.
- **Strategic Architectural Leadership:** Formally championed **Azure Lighthouse** over Azure Managed Applications to resolve cross-tenant visibility constraints; authored deep-dive technical evaluations demonstrating how Lighthouse's automation via Azure Marketplace would have accelerated time to market and eliminated multi-tenant architectural debt for **ALL of Azure Global**.
- **Identity & Authentication:** Streamlined the SMS-based login flow by implementing custom Identity Server solutions over Azure Cosmos DB and collaborated with Telesign to significantly reduce SMS code-sending costs across many countries; Engineered workflows for load-testing authentication flows using locally signed JWT tokens.
- **Search Infrastructure & Optimization:** Designed and implemented a custom search index featuring advanced schema optimizations and Lucene-based retrieval; Used Azure Cognitive skills to index relevant content from unstructured file formats like PDF, Office, Videos, Audios, etc. Used Azure Cognitive Search to improve search relevance with pretrained semantic analyzers like fuzzy matching, edit distance, synonyms, phonetics, etc.

- **Media Security & DRM:** Authored a technical proposal to modernize content protection for temporal media using industry-standard DRM protocols like Digital Rights Management including **Widevine, PlayReady, and FairPlay** beyond just AES-128 based encryption-in-transit.
- **Notifications Infrastructure:** Onboarded via client-side Xamarin APIs before assuming full ownership of the platform's notification subsystem (under the mentorship of [Rohan Badlani](#))
- **High-Throughput Bulk Ingestion:** Architected and built an end-to-end bulk upload system enabling parallel ingestion of multi-format media; significantly reduced administrative onboarding time and **super-charged the manual testing experience** for everyone.
- **Developed a scalable monitoring infrastructure** using Event Hubs and Azure Data Explorer to centralize telemetry and enable diagnostic and analytic dashboards.
- **Automated end-to-end deployments and CI/CD pipelines** enabling Public Preview release of the offering on Azure's Marketplace.
- **Advanced Thread Modeling:** Conducted deep-dive threat modeling and attack surface analysis using Microsoft internal security frameworks; ensuring the integrity of user data and platform content.
- **Privacy & Compliance:** Enabled automated compliance tooling to automate the identification of secure, licensed dependencies; ensuring rigorous adherence to global data protection and privacy standards.
- **State Persistence & User Analytics:** Developed features for real-time tracking of user progress, including "most recently viewed lesson" and "last watched timestamp" for temporal media content ensuring a seamless cross-device learning experience.



Microsoft Hackathon

Engineer

2018 — 2020

Hyderabad, Telangana, India

- **Kool AI(d) for Gamers:** Fine-tuned pre-trained computer vision models to extract real-time HUD metrics and game states from live Dota 2 streams; Collaborated in a team of two.
- **Koev Halev:** Actively contributed at the Microsoft IDC campus, providing domain expertise in cross-functional design discussions for a project focused on emotional well being and mental health support.



Microsoft Code.Fun.Do

Engineering Lead and Pizza Logistics Moderator (supervised by [Sandeep Chadda](#) & [Vijay Vavilala](#))

2018 — 2019

India

- **Event Moderator:** Delivered talks on project-submissions, technical-guidelines etc. and managed ad-hoc troubleshooting and Pizza logistics at **BITS Pilani (Hyderabad)** and **NIT Surathkal**.
- **Competitive Programming:** Proposed supplementing project-based events with structured programming contests (Google Kickstart, Google Code Jam, Meta Hacker Cup, ACM-ICPC, etc.), positioning rigorous algorithmic proficiency as a **foundational pillar for engineering talent**.



The Garage, Microsoft

Engineer

June 2018

Hyderabad, Telangana, India

- **IoT Home Automation:** Collaborated within a team to retrofit and control legacy electric switches with motorized 3D-printed actuators via a Raspberry-Pi based IoT layer.



Goldman Sachs

Engineering Lead

September 2016 — October 2016

Bengaluru, Karnataka, India

- **National Runner-Up (GS Quantify 2016):** Led a 3-member team in this flagship interdisciplinary event, serving as the primary contributor for the advanced algorithmic problem-solving track.
- **Data Science** (ft. [Samiksha Gupta](#)): Contributed to the data-science track for bond-liquidity-prediction.
- **Quant** (ft. [Shivam Tyagi](#)): Directed the quantitative solution track to finalize the submission.
- **Executive Presentation:** Earned a fully sponsored invitation to their Bengaluru headquarters (hosted at the prestigious Hilton Embassy) where the team delivered the concluding presentation.



Microsoft Foundry

Engineering Internship (supervised by [Pankaj Garg](#) & [Saumya Chandra](#))

May 2016 — July 2016

Hyderabad, Telangana, India

- **Conversational AI:** Engineered chat-bots using the **Microsoft Bot Framework**, architecting custom conversational flows for automated form-filling, within a cross-functional team.
- **Computer Vision:** Processed images using **OpenCV** to better highlight textual information, significantly improving downstream OCR engine performance.
- **Natural Language Processing:** Tested and evaluated experiences built with **Microsoft LUIS**.
- **Live Technical Demonstration:** Delivered a final product presentation to a large audience, showcasing the templated conversational flows for automated form-filling.



Xerox Research Centre

Research Intern (supervised by [Vaibhav Rajan](#))

Dec 2015 — Jan 2016

Bengaluru, Karnataka, India

- **Xerox Research Innovation Challenge:** Engineered predictive neural network models, augmenting the provided patient vitals with synthetically generated data to predict mortality of ICU patients.
- Completed a Winter School on predictive data-science (ft. [Dushyant Kumar](#) & [Sartaj Singh](#))
- Preprocessed massive unstructured lab-result datasets and built ensemble models combining them with patient vitals to predict both mortality and broader health complications.
- Secured **third place** (online-round) and **seventh place** (onsite-round) in the overall challenge; successfully competing against specialized research teams.

ACCOMPLISHMENTS

- **ACM-ICPC 2016** (ft. [Umang Goel](#) & [Himadri Mishra](#)): Represented IIT (BHU) Varanasi at the ACM-ICPC Asia-Amritapuri regionals.
- **ACM-ICPC 2015** (ft. [Aditya Kapoor](#) & [Shivam Tyagi](#)): Represented IIT (BHU) Varanasi at the ACM-ICPC Asia-Amritapuri regionals.
- Ranked **15th** internationally (2nd in India) in **Codeforces Round #410 (Div. 2)**
- Ranked **83rd** internationally in **Google Kickstart Round D 2017**
- Ranked 16th on HackerEarth's October Sky among 800 international candidates
- Ranked 25th in IIT Bombay's TechFest-15 International Coding Challenge Final Round
- Ranked 12th in IIT Bombay's TechFest-15 International Coding Challenge Screening Round
- Ranked **19th nationally** in CodeChef May Challenge 2015, reaching an **overall India Rank of 31** in CodeChef's Long Challenge track.
- **IIT Kanpur's Monthly Programming Contest:** Ranked 41st among 1117 other teams (ft. [Abhishek Bind](#) & [Rishabh Daal](#))
- **IndiaHacks 2016** (ft. [Sartaj Singh](#)): Secured a **14th rank** against 3,000+ other teams in their data-science track; Invited to the on-site finals held at **Vivanta by Taj** in **Bengaluru, KA**

PROJECTS

- **2D Physics Game Engine:** A gravitational rigid-body simulation in Java:
 - Engineered a physics engine featuring multi-object **collision detection and kinematic interactions**
 - Engineered a **Near-Earth fixed-gravity model** to [simulate realistic downward acceleration](#).
 - Engineered a **Newtonian $1/r^2$ gravity model** to simulate **N-body interactions** and **galactic mergers** in empty space.
- **Multi-Agent Environment Simulation:** Built a simulation of independent RL-Agents coordinating with each other to efficiently collect items distributed in a two-dimensional grid.

- **AIMA-CSharp Open Source Contribution:** Contributed to the [AIMA-CSharp repository](#) by implementing pseudo-code in C# from Peter Norvig's *Artificial Intelligence: A Modern Approach*.
- **Image Steganography:** Implemented a system to encrypt and embed hidden messages within images.
- **Intel E5200 CPU Overclocking:** Successfully overclocked an Intel Pentium E5200 processor from a base clock of **2.5GHz** to **2.8GHz**. Involved tuning **FSB frequencies, CPU multipliers, and voltage settings**; Benchmarked thermal output and system stability under peak loads.
- **The Chess Game:** Developing an evolving chess engine from pre-college through my final year, transitioning from a multiplayer foundation (featuring UDP, TCP, etc.) in Java to a sophisticated AI-driven system. Independently implemented an AI engine utilizing **Mini-Max** and **Alpha-Beta pruning**.
- **Tic-Tac-Toe AI:** Implemented a decision-tree based AI and multi-player for Tic-Tac-Toe; the computer being unbeatable in the hard difficulty level.

SKILLS

- **Leadership** — Prioritization, Cross-functional Collaboration, Reviews, Model, Coach, Care, Ally, etc.
- **Artificial Intelligence** — Reinforcement Learning (Agentic Workflows, Exploration vs Exploitation, Play, Reward Optimization, etc.), Deep Learning (Neural Networks, CNNs, RNNs, Transfer Learning, etc.), Machine Learning (Mathematical Modeling and Search Heuristics), Data Science (Statistics, Classification, Regression, PCA etc.), LLM Applications, etc.
- **Engineering** — Data Structures and Algorithms, Distributed Systems and High Performance Computing (Billions of daily notifications), Data Pipelines, Authentication and Identity Systems, Blockchain, API design and Smart Contracts, IaaS, etc.
- **Site Reliability** — Business Continuity and Disaster Recovery, On-call and Incident Response etc.
- **Data & Storage** — Redis, NoSQL, SQL, Blockchain, etc.
- **Search** — Apache Lucene, Elasticsearch etc.
- **Cloud & Infrastructure** — Microsoft Azure (Expert), AWS, Kubernetes, CDN, CI/CD & Infrastructure Automation, etc.
- **Frameworks & Libraries** — JVM, .NET, TensorFlow, Keras, Pandas, Scikit-learn, XGBoost, OpenCV, Unity, React, ARM, IronPDF, etc.
- **Hardware** — Semiconductors, Assembly, Firmware, etc.
- **Security** — Ethical Hacking, Capture the Flag events, Threat Modeling, etc.
- **Privacy** — Encryption, CCPA, GDPR, etc.
- **Programming Languages** — Logo, Basic, Java, C, C++, C#, Python, TypeScript, PowerShell, Hack, etc.
- **Natural Languages** — English, Hindi, Marwari, Sanskrit, etc.

EDUCATION



Indian Institute of Technology (Banaras Hindu University)

July 2013 - May 2018

Integrated Master's degree (5 years): Mathematics & Computing

Varanasi, Uttar Pradesh, India

- **Thesis:** Deep Learning based Classification of Handwritten Mathematical Symbols.
(Supervised by [Tanmoy Som](#))
 - Trained and fine-tuned transfer learning models to achieve state-of-the-art accuracy on the HasyV2 dataset.
 - Successfully defended thesis through a detailed technical presentation on CNNs, transfer-learning methodologies and classification results.
 - Technologies Used: Pandas, AWS, Numpy, Tensorflow, Keras, CNN, MobileNet, DenseNet, ResNet, Wide-ResNet, NASNet, Scikit-learn, etc.

- **Teaching Assistant** for the freshman course on Mathematical Methods.
- **Problem Setter & Moderator – Hackerrank (Technex'16):**
 - **Mathletics:** www.hackerrank.com/mathletics-technex16
 - **Prayaas:** www.hackerrank.com/prayaas-technex16
- **Ranked 1st in CodeDef-2016**, a coding contest organized under Udyam, a college festival organized by the Electronics branch.
- **Ranked 2nd in Analiticity**, a campus-wide data-science contest in Technex '16.
- **Ranked 1st in COPS-OPC-2015**, a programming contest organized by the club of programmers.
- **Microsoft Code.Fun.Do** Collaborated in a team of 4, developing a 2D top-down space shooter game using the Unity game engine.
- **Matheletics:** Ranked 3rd in the Mathematics and Computing event, in Technex-15.
- **Predicting Tourism Trends:** Participated in a team of 3 to build machine learning models to forecast tourism trends (Supervised by [Santwana Mukhopadhyay](#))
- **Rediscovering Darkness:** Stood 2nd in an Astronomy Quiz hosted by the Astronomy Club!
- **KashiYatra-14:** Runner's up in Treasure Hunt, participated in a team of 3.
- **The Chess Game:** Stood 2nd in Technex-14's open-source Mod-Ex event.
- **Robo-Soccer:** Runners-up out of 100+ teams at the Robotics-Club's Technex-13 event; conceptualized electronic circuit designs for a D-Pad controller to operate an All-Wheel-Drivetrain.
- **Event Organizer & Moderator:** Competitive tournaments for Age of Empires, Rise of Nations, Call of Duty, Need for Speed, and Defense of the Ancients!

Coursera.org

Specializations and Certifications

April 2018 — March 2025

Online

- **DeepLearning.AI:** [Deep Learning Specialization](#)
- **University of Alberta & Amii:** [Reinforcement Learning Specialization](#)
- **The State University of New York:** [Blockchain Basics and Smart Contracts](#)
- **DeepLearning.AI:** [Generative AI For Software Development](#)

Forum for IIT-JEE

Joint Entrance Examination

April 2011 — June 2013

Dhanbad, Jharkhand, India

- **All India Rank 494** in Joint Entrance Examination — 2013, among 1.3 million candidates.
- **All India Rank 1672** in IIT-JEE (ADVANCED) — 2013, among 150k candidates.
- **Stood 1st** in the batch of 2013, among 200+ other candidates.



Delhi Public School

Central Board of Secondary Education

June 2011 — April 2013

Dhanbad, Jharkhand, India

- **Scored 96.5%** in the All India Senior School Certificate Examination (AISSE) — 2013!
- **All India Rank 6355** in National Eligibility cum Entrance Tests (NEET) — 2013, among 650k candidates
- **Awarded DST INSPIRE Science Scholarship for ranking in the top 1%** of AISSE — 2013
- **State Rank 1 (Medicine)** in Jharkhand Combined Entrance Competitive Examination — 2013
- **State Rank 2 (Engineering)** in Jharkhand Combined Entrance Competitive Examination — 2013
- Recognized in the **top 0.1% of AISSE — 2013 for excellence in Physical Education**
- Drafted into the advanced mathematics program **Aryabhata** and represented the school at IMO!
- **Labs:** Biology, Chemistry, Physics!



- **Received 90%+ in FTRE Scholarship** based on competitive national-level performance among 100,000+ candidates!
- **Scored 87.9 %** in the Indian Certificate of Secondary Education — 2011 examination.
- **Practicals:** Reading, Speaking, Writing, Physical-Training, Art & Craft, Computers, Chemistry, Biology, Physics, Topology, etc.

PURSUIITS

- **Metaverse** — Spatial Computing, AR/VR, WIP!
- **Science** — Medicine, Anatomy, Physics, Chemistry, WIP!
- **Music** — Rock, Pop, EDM, MeaningWave, LoFi, Rap, etc.
- **Culinary** — Foodie (Expert), Cooking (Moderate)
- **Fitness** — Swimming (Beginner), Martial-Arts (Beginner), Gym (Moderate), Yoga, Badminton, etc.
- **Multiverses** — RPG, RTS, FPS, Podcasts, Documentaries, [Books](#), Anime, WIP!
- **Traveling** — Stars, NASA, SpaceX, Amazon Leo, Auroras, Universities, Museums, Monuments, etc.
- **Philosophy** — Ontology, Epistemology, Causality, Agency, Teleology, Teleonomy, etc.
- **Natural Languages** — Mandarin, Spanish, Japanese, Punjabi!
- **Guitar** — Beginner (Eartuning, Staff Notation, Electric, Yousician, etc.)